



NETFLIX¹

Company Overview

Netflix was founded in 1997 and was conceptualized as the first online video rental option in the United States. Netflix launched its online DVD rental service in September of 1999. Today, Netflix remains the largest online movie rental service providing more than 3.8 million subscribers access to a comprehensive library of more than 35,000 movie titles. The standard Netflix package allows subscribers to have up to three titles out on rental at a time with no due dates, late fees or shipping charges. This basic service is priced at \$17.99 per month. In addition to this basic service, Netflix offers plans with different price points that allow subscribers to keep either fewer or more titles out at a time. In 2005, the average paying Netflix subscriber paid \$18.35 for the service (blended across all of Netflix product offerings).

Each Netflix user maintains a queue of titles they would like to view. When one title is returned, the company's systems automatically scan the user's queue to determine the next available title. That title is then packaged, posted and sent to the user's home. In this fulfillment operation, Netflix utilizes a network of forty distribution centers and ships more than 700,000 DVDs each day. Netflix's historical financials are summarized in exhibit 1.

Revenue Growth

Netflix's revenue is driven by two basic forces: number of customers and average price per subscribing customer.

Netflix management thinks of the size of its customer base as a function of Netflix's penetration of US DVD households. At the end of 2005, there were 112.3 million households in the United States. The US Census bureau expects the number of households to grow predictably at 90 bps (0.9%) per year. Of these households ~98.5% owned a television. Industry experts estimate that the television is now completely penetrated and expect this 98.5% penetration rate to remain consistent for the foreseeable future. Of television households, 67.8% owned a DVD player by the end of 2005. Exhibit 2 provides both historical and projected DVD penetration rates. In 2005, the net of these macro factors was a 75 million household base into which Netflix could market its product.

¹ Ryan N. Cotton and Jim R. Hart III, MBAs 2007, prepared this case under the supervision of Professor Peter DeMarzo, the Mizuho Financial Group Professor of Finance, as the basis for class discussion rather than to illustrate either effective or ineffective handling of an administrative situation.

This case is based on publicly available information. Any guidance or non-public information is strictly fictitious. Certain changes have been made to actual Netflix accounting policies and expense categories in the interest of simplicity.

Of these 75 million households, Netflix has been able to successfully sell its service to ~5.1% or 3.8 million customers. This penetration is the result of a number of factors including direct marketing, word-of-mouth marketing, referral bonuses, co-marketing arrangements with consumer electronics manufacturers and retailers and the rollout of Netflix distribution centers, which allows Netflix to access an increasingly broad customer base. Looking forward, Netflix management believes it will be able to continue the expansion of its customer base. Management has outlined three scenarios for its future growth. These include considerations of the marketing budget, competitive threats and planned DC rollout:

- Base Case – Assumes geographic penetration of new markets in accordance with Netflix expansion plans. New markets are expected to ramp in line with Netflix experience, culminating in >7% penetration by year four (as was achieved in San Francisco – see exhibit 3 for an overview of Netflix’s historical new market experience).
- Aggressive Case – Assumes same geographic expansion as base case, but assumes Netflix penetrates new markets at a faster rate. This rate is derived by extrapolating the success to date of the markets Netflix entered in 2002 and 2003.
- Conservative Case – Assumes faster penetration of video on demand (both in cable TV and online versions) forces Netflix to delay or abandon geographic expansion. Growth is driven by continued ramp of existing markets, adjusted downward for cannibalization impact of VOD.

These results of these projections are outlined in Exhibit 4.

In addition to new customers, Netflix must work each year to replace customers that have discontinued using the service. Netflix measures these customer losses as “churn” and then reports this information to the street in the form of “monthly churn rate.” Netflix defines this metric as follows:

$$R = \frac{C}{A * 12}$$

Where,

R: Monthly churn rate expressed as a %;

C: Total number of churned customers in a year

A: Average numbers of customers in a year ((# at beginning + # at end) / 2)

Alternatively, this expression can be rewritten to derive total churned customers. These can be expressed by taking the average number of customers in the year (last year’s end of year total plus this year’s end of year total divided by two) and multiplying by twelve times the churn rate.

Combining the total churned customers and the net adds in the period (this year’s end of year customer base less last year’s end of year customer base), yields the gross number of customer adds over the period: the number of customers Netflix had to acquire to both replace those it lost and to grow (total added customers = # at end of year - # at beginning of year + # churned).

As an example, in 2005, Netflix reported monthly churn of 5.12% and 3.84 million subscribers at year end. In 2004, Netflix reported 2.61 million subscribers at year end. This implies that Netflix

churned 1.97 million customers in 2005 $((3.84+2.61) / 2 * 12 * .0512 = 1.97)$. Since Netflix added 1.23 million *net* customers in 2005 and churned 1.97 million existing customers, Netflix acquired 3.2 million gross new customers over the year.

Netflix management has been able to reduce churn rates steadily over the past four years, from a high of 6.9% in 2002, to 5.4% in 2003, 5.2% in 2004 and 5.1% in 2005. Management believes that churn will stabilize at 5% per month, beginning in 2006.

Finally, some of Netflix's customers are acquired under a free trial arrangement. The net effect of this arrangement is that 96% of Netflix's customers at any given point are paying customers. Management believes this relationship between paying and non-paying customers will remain relatively stable in the future. Of those that do pay, each provides \$18.35 per month in subscription revenue. Management expects pricing to be flat going forward, although there is some concern that increased competition may force Netflix to lower prices.

On occasion, Netflix will sell DVDs to its subscribers. Revenue from such sales totaled ~\$6.2 million in 2005. This revenue should grow in proportion to Netflix's subscriber base. Netflix earned a 39% gross margin on these sales in 2005 and expects to be able to maintain this margin going forward.

Cost of Sales

Netflix's cost of fulfilling customer rentals is essentially driven by the customer's utilization of the service and can be thought of as having four basic components: library amortization costs, revenue sharing costs, postage cost and packaging costs.

- Library amortization costs – Netflix purchases a number of titles to extend the breadth and scope of its video library. The cost of new release acquisitions is amortized over a one year period. The cost of catalog, or 'classic,' acquisitions is capitalized and amortized in an accelerated fashion over a three year period [Exhibit 5 outlines the logic and mechanics behind this policy]. In 2005, Netflix recorded an amortization expense of \$96.9 million related to its film library. As Netflix plans to continue to expand this library, management estimates that this cost will grow at roughly 8% per year, but has provided a more detailed schedule of anticipated amortization expense (in keeping with Netflix accounting policies) in exhibit 6.²
- Revenue sharing – In addition to titles shipped from its library, Netflix ships the majority of titles under revenue sharing agreements with major film studios. Under these agreements, Netflix is saved the capital risk of purchasing the DVDs upfront but can, instead, share revenues garnered from their utilization with the studios after the fact. In 2005, 70% of Netflix's disk shipments were covered under these agreements and

² In this case, the work of converting CapEx to amortization expense using an accelerated depreciation schedule has been done for you. As you will be asked to build a depreciation schedule in future models, it may be helpful to use this as a reference. Implicit in this schedule are the assumptions that a) all CapEx occurs at the beginning of the fiscal year, b) 20.4% of purchases are new releases that are amortized in the first year and c) catalog releases are amortized 50% in year one, 35% in year two and the balance in year three.

management believes this proportion will remain constant in the future. For each disk shipped under such an arrangement in 2005, Netflix remitted \$1.04 to a studio. The contracts governing these relationships include built in price escalators. As such, the cost per disk shipment should grow at 2.5% per year (roughly in line with inflation).

- Postage – For each disk shipped, Netflix must pay postage (\$0.40 per disk in 2005). This should grow with postage rates at 2.5% per year.
- Packaging – Netflix ships each disk in a specially designed mailer that cost \$0.05 each in 2005. As with postage, this cost is expected to grow at an inflationary 2.5% per year.

In 2005, Netflix customers received an average of 6.5 disks per month. This utilization rate is expected to remain constant going forward.

Overhead

Netflix divides its overhead into five basic categories: marketing, fulfillment, technology & development, G&A and stock based compensation expense.

- Marketing – Marketing expense reflects the cost of acquiring customers. Netflix reports subscriber acquisition cost (“SAC”) on an annual basis. In 2005, Netflix reported an average SAC of \$44.34 per *gross* subscriber added. This is expected to grow at 2.5% per year as Netflix faces increased competition and is forced to dig deeper into the potential customer base to find new subscribers.
- Fulfillment – This expense represents the essentially fixed component of costs related to pick, pack and ship to customers including, but not limited to, depreciation of distribution center facility expenses and labor to staff the DCs. Netflix management expects this expense to grow at 10% per year in 2006 and 2007, driven by new DC openings and wage rate inflation, then slow to 2.5% in 2008.
- Technology & Development – This expense includes the cost of developing, improving and maintaining both the Netflix website and the company’s order fulfillment systems. Management estimates that, in 2005, this expense was roughly 50% fixed and 50% variable (obviously, this ratio will change over time). Netflix’s IT plans call for the variable component to remain stable at 2.2% of sales and the fixed component to grow at 5% per year through 2008, then level off at 2.5% in 2009.
- G&A – Includes all basic corporate overhead functions such as finance, legal, administration and strategic planning. Management estimates that, in 2005, 70% of this expense was fixed and 30% was variable with revenue. The variable component is expected to remain constant at ~1.6% of sales while the fixed component is expected to grow at 5% per year in 2006 and 2007 as Netflix completes its national rollout and is expected to level off to grow with inflation (2.5% per year) beginning in 2008.

Other Information

Netflix is a debt free business and does not anticipate any borrowings in the future. The business does, however, earn interest on the cash balance necessary to operate the business. This necessary cash balance is assumed to remain constant over the projection period. Management has provided a schedule of anticipated interest income in exhibit 7.

Due to the billing and payment cycle, Netflix operates with negative working capital. It collects from its subscribers at the beginning of each month and does not remit payments to vendors such as major film studios for at least 60 days. In 2005, net working capital was -9.2% of sales. Management expects net working capital to stabilize at -9.0% of sales in 2006.

Netflix had 62.6 million shares outstanding in 2005. Management has prepared a schedule of the anticipated evolution of this share count (taking in to account new option issuances). This schedule can be found in exhibit 7.

Historically, Netflix has not been a taxpayer. This is the result of significant net operating loss carry-forwards generated in the early days of the business. For the purposes of creating financial projections, these tax assets should be ignored and Netflix's forward earnings should be subjected to the statutory corporate tax rate of 35%. The value of these tax assets will be considered as part of the valuation (either in an APV analysis or a sum of the parts valuation) in the future.

Netflix anticipates non-library acquisition CapEx to be equal to depreciation expense. A schedule of library acquisition CapEx is provided in exhibit 7.

Assignment

The time is early 2006³, and you are a public equity analyst at *MBA First National* covering the entertainment sector. Your firm has recently initiated coverage of Netflix. Given the management assumptions outlined in the case (gathered from analyst calls, meetings with management and your industry expertise), prepare a pro-forma model of Netflix's earnings from 2006 to 2010 and a presentation you will use to explain this model to your clients.

In your analysis, there are several key sensitivities you should outline as investors are particularly concerned about Netflix's exposure to several potential competitive threats and market trends.

Part I: Model Output

You should prepare and submit a model that contains the following:

- 1) A full pro-forma income statement for Netflix from 2005A to 2010E. This should include both an EBIT and EPS forecast under the "base" case for Netflix market penetration.
- 2) EPS estimates, by year, from 2006 to 2010 for Netflix in each of the three penetration cases: "Base," "Aggressive" and "Conservative"
- 3) A "6 line cash flow statement" (remember, this tool is designed to derive Cash Flow for Financing Activities), a Free Cash Flow (FCF) calculation and a Free Cash Flow to Equity (FCFE) calculation for each year of the "base" case. We will cover different calculations and definitions of cash flow more in class, but, in this case, follow the outline below:

EBITA (EBIT + Library Amortization Expense)⁴

Plus: Interest Income (Expense)

Less: Taxes

Less: Change in Working Capital

Less: Library Acquisition CapEx⁴

= Cash Flow for Financing Activities (CFF)

Cash Flow for Financing Activities

+ After-Tax Interest Expense

= Free Cash Flow (FCF)

Cash Flow for Financing Activities

+ Increase in Net Debt

= Free Cash Flow to Equity (FCFE)

³ This implies that 2005 will be the last year of actual / historical data and 2006 will be the first year of projections in your model.

⁴ Typically, a 6 line cash flow will start with EBITDA, but as we do not know depreciation and we are assuming that CapEx (excluding library acquisition costs) is equal to depreciation, we can start with EBITA and include only the library CapEx in the cash flow statement (*be sure you understand how and why this works as it will be a critical in future models*)

- 4) 2010E Cash Flow for Financing Activities (“CFF”) numbers for the following sensitivities (all under the “base” case penetration scenario):
- Churn rate – 3%, 5% and 7.5% each year from 2006 to 2010
 - Utilization – 5.5 / Month, 6.5 / Month and 7.5 / Month
 - Pricing growth (monthly subscription revenue / customer) – 0%, -5.0% and -8.5% growth per year
 - Subscriber acquisition cost (“SAC”) – 2.5%, 5% and 10% growth per year

Hints / Suggestions for the Model

- Build your model with as much flexibility as possible, this will make the sensitivities easier
- Label key assumptions separate from the income statement – treat the income statement as output, not the guts of your model; and use the historical income statement as a guide
- Buildup key operating assumptions / metrics first – a few suggestions:
 - o Start with households – # in US, # with TV, # with DVD players and # with Netflix (from penetration curves and case guidance)
 - o Next, calculate the relevant customer statistics:
 - Customer count: beginning of year & end of year (beginning of year + net adds)
 - Change: net adds & gross adds (remember, gross adds = net adds + # churned)
 - Other metrics: churned customers (note: this will be a very high number – if Netflix is churning 5% of its customers a month, this will be ~70% of the customer base – also, as this is a monthly figure, you should calculate total churned customers in the year off of average number of customers in the year), paid subscribers (96% * average subscribers in year), etc.
 - o Use these calculations to build up subscription revenue; model sales revenue w/ guidance in case
 - o Calculate service utilization (DVDs shipped per month) as this is a critical cost driver
 - o Other expense items should be modeled in accordance with the guidance in the case

Part II: Presentation

In addition to the model, you should also prepare and submit a presentation that answers the following questions.

[Note: In future cases, less guidance will be provided on the layout of the presentation and you will have more flexibility around and authorship of the layout and content. However, as this is our first case, it may be helpful to have a shared template. As such, please follow the outline below.]

Slide 1: Model Results

- EPS forecast under each of the three operating scenarios (aggressive, base and conservative)
(this may be presented graphically or in Excel form)
- A 6 Line Cash Flow statement for the ‘base’ case
- In presenting this slide, be prepared to speak to critical assumptions made in the model, primary drivers or the results and areas where assumptions may be aggressive or conservative. The goal of this slide is to orient potential investors to trends in NFLX’s financial results and areas of potential concern or opportunity. You may find it helpful to ‘flag’ or highlight some of these issues or trends in the slide itself.

Slide 2: Sensitivities

- Present the results of the sensitivity analysis conducted in question 4 above
- Attempt to answer (both on the slide and in your presentation of that slide) the following questions:
 - o What are the key metrics that drive NFLX's performance and on which investors should focus (i.e., is lowering churn more or less important than reducing customer acquisition cost?, etc.)
 - o What scenarios should investors fear? What scenarios present upside opportunities for the company?
 - o Given the facts of the case and your knowledge of the market and Netflix's service, what do you believe to be the most likely set of results and the most likely 2010 cash flow picture for Netflix?

Slide 3: Looking Forward

In 2006, Netflix management announced an aggressive and bold move into online video delivery. This service is intended to compete with both video on demand ("VOD") and other digital download services (such as iTunes). This program has the potential to open up an entirely new business line for Netflix, but requires substantial capital investment (to construct a viable download platform and digital storage and service equipment).

- Thinking strategically, do you believe this is a wise move? As an analyst covering the stock, how do you position this decision with investors (i.e., is this something they should be excited about or something they should be leery of?)
- If you were to adjust your model to incorporate this decision, reflect on the following:
 - 1) What additional data would you want?
 - 2) What structural changes would you make to the model itself?
 - 3) What additional sensitivities, if any, would you run?

Exhibit 1 – Netflix Historical Financial Performance

(All Numbers in '000s)	Historical			
	2002	2003	2004	2005
Revenue				
Subscription	150,818	270,410	500,611	681,779
Sales	1,988	1,833	5,617	6,215
TOTAL Revenue	152,806	272,243	506,228	687,994
YoY Growth %	89.3%	78.2%	85.9%	35.9%
Cost of Sales				
Cost of Sale Merch	1,092	624	3,057	3,794
% of Sales Rev	54.9%	34.0%	54.4%	61.0%
Library Amortization	17,417	43,125	80,346	96,883
Revenue Sharing Costs	28,509	58,973	113,164	183,106
Postage Costs	29,149	41,489	71,902	100,608
Package Costs	1,970	4,149	7,989	12,576
Cost of Subscription	77,044	147,736	273,402	393,172
% of Subscription Rev	51.1%	54.6%	54.6%	57.7%
TOTAL Cost of Sales	78,136	148,360	276,459	396,966
Gross Margin	74,670	123,883	229,770	291,028
Gross Margin %	48.9%	45.5%	45.4%	42.3%
Operating Expenses				
Marketing Expense	35,783	49,949	98,027	142,006
% of Revenue	23.4%	18.3%	19.4%	20.6%
Fulfillment Expense	19,366	31,274	56,609	70,762
% of Revenue	12.7%	11.5%	11.2%	10.3%
Tech & Dev Costs	14,625	17,884	22,906	30,942
% of Revenue	9.6%	6.6%	4.5%	4.5%
G&A Expense	15,569	20,304	32,874	35,867
% of Revenue	10.2%	7.5%	6.5%	5.2%
Other Charges / Act. Adj.				3,200
TOTAL Operating Expenses	85,343	119,411	210,416	282,777
% of Revenue	55.9%	43.9%	41.0%	41.1%
EBIT	(10,673)	4,472	19,353	8,251
EBIT Margin %	-7.0%	1.6%	3.8%	1.2%
YoY Growth		N/A	332.7%	-57.4%
Interest Income (Expense)	(10,275)	2,040	2,422	5,753
Pre Tax Income	(20,948)	6,512	21,775	14,004
Income Tax	-	-	181	-
Effective Tax Rate				
Net Income (Loss)	(20,948)	6,512	21,594	14,004
Net Income Margin %	-13.7%	2.4%	4.3%	2.0%
Shares Outstanding	28,204	62,884	64,713	62,596
Earnings Per Share (EPS)	\$ (0.74)	\$ 0.10	\$ 0.33	\$ 0.22
<u>Memo:</u>				
Library Acquisition CapEx	(24,070)	(55,620)	(102,971)	(113,950)

Exhibit 2 – DVD Penetration

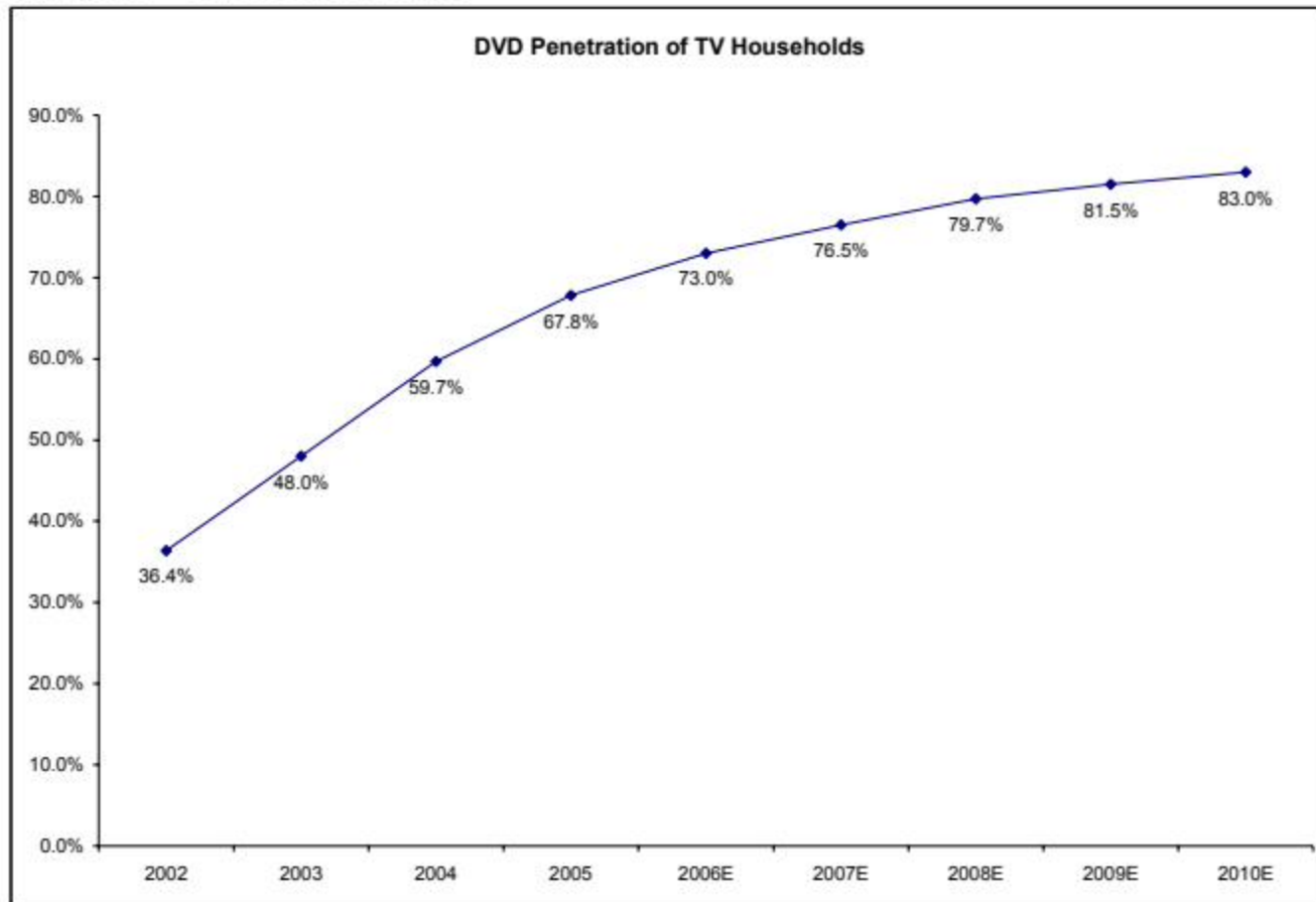


Exhibit 3– Historical Netflix Market Penetration Experience

Netflix Market Penetration

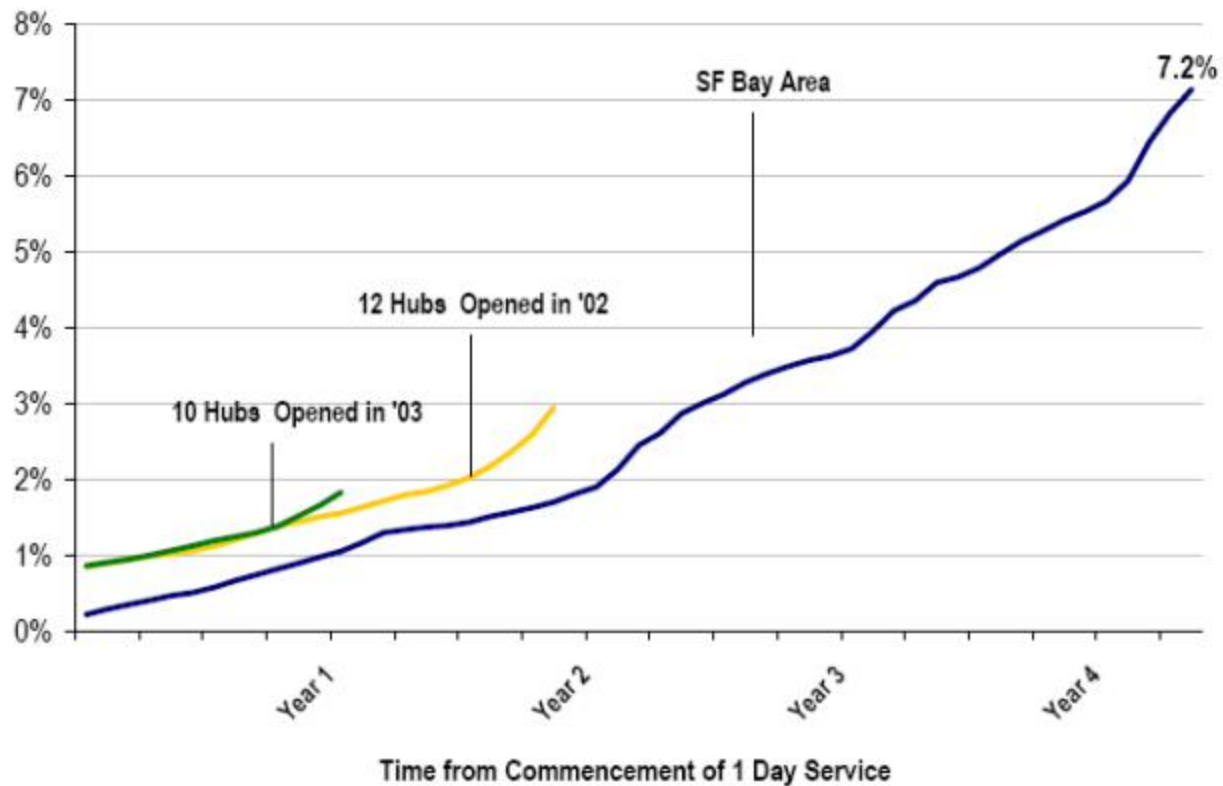


Exhibit 4 – Projected Netflix Penetration

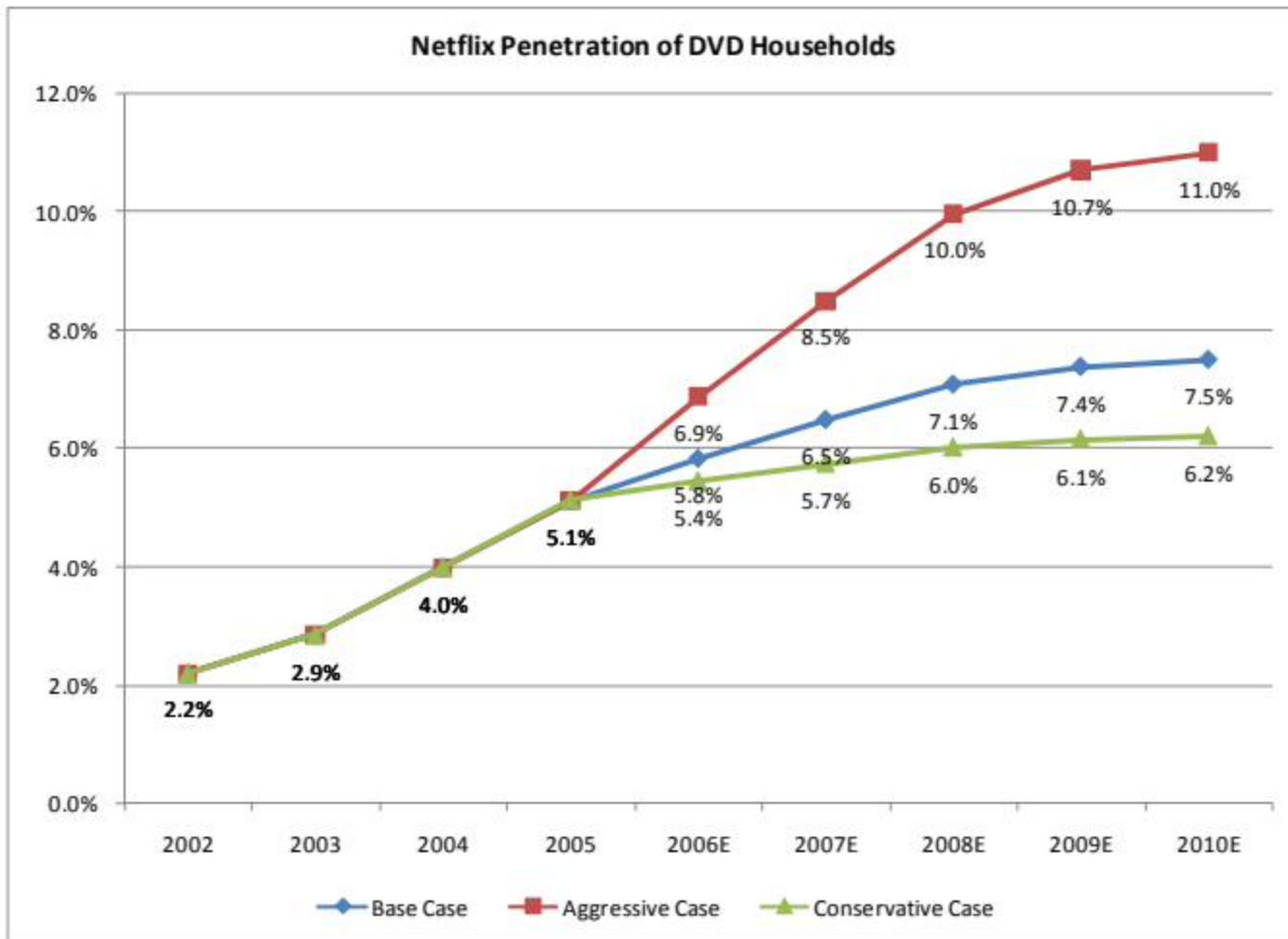
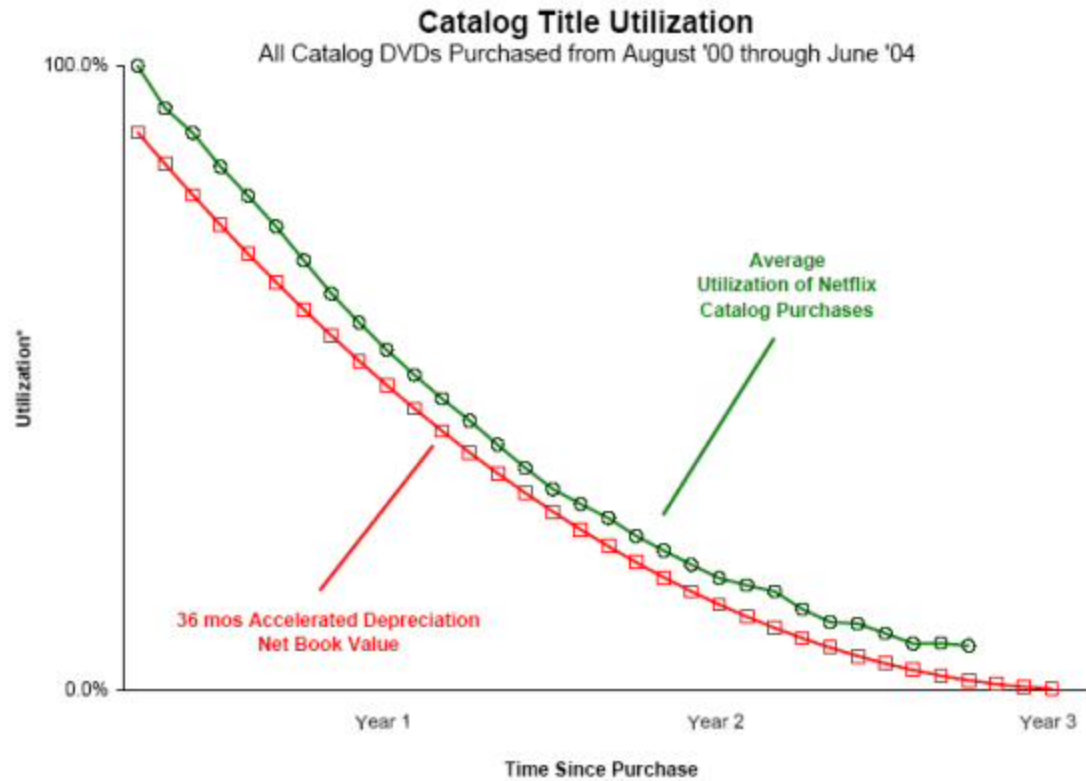


Exhibit 5 – Catalog Utilization and Library Amortization



*Utilization of Netflix catalog purchases is calculated by dividing the shipments in each month by the quantity initially purchased and indexing the resulting value to the ratio of the first full month's shipments as a % of the initial quantity purchased.

Exhibit 6 – Anticipated Library Amortization Expense

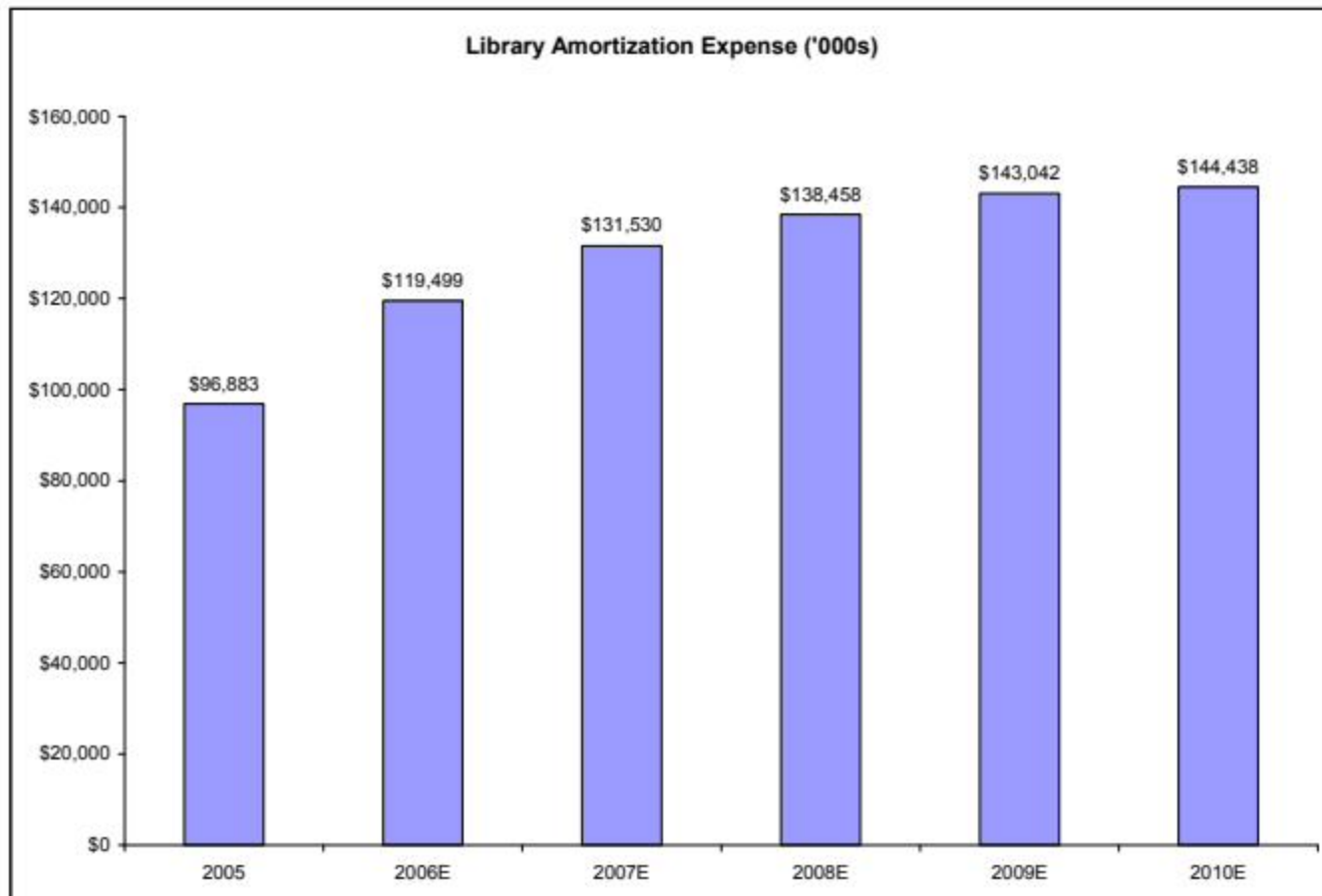


Exhibit 7 – Supplementary Management Schedules

<i>(all numbers in '000s)</i>	2005A	2006E	2007E	2008E	2009E	2010E
Interest Income (Expense) ¹	5,753	5,753	5,753	5,753	5,753	5,753
Shares Outstanding	62,596	67,831	69,571	71,310	73,050	74,789
Library Acquisition CapEx	113,950	125,345	137,880	141,326	144,860	144,860

¹ Reflects interest on cash necessary to operate the business.